Commutative Algebra & Algebraic Geometry

Exercise 6

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6.1 Problem

Fix a morphism of rings $R \to A$.

(1) Assume that A is a flat R-module, and let M be a flat A-module. Show that M is flat as an R-module.

(2) Let M be a flat R-module, show that $A \otimes_R M$ is a flat A-module.

(3) Let S be a multiplicative subset of R, and let R be a flat R-module. Show that $S^{-1}M$ is a flat $S^{-1}R$-module.

(4) We say a morphism of affine schemes is **flat** if the corresponding morphism of rings is flat. Show that $\mathrm{Spec}(A) \to \mathrm{Spec}(R)$ is flat iff there exists an open cover $\{\mathrm{Spec}(R)_{f_i}\}_{i\in I}$ of $\mathrm{Spec}(R)$ such that the corresponding maps $\mathrm{Spec}(A) \times_{\mathrm{Spec}(R)} \mathrm{Spec}(R)_{f_i} \to \mathrm{Spec}(R)_{f_i}$ are all flat.

(5) Suppose $0 \to M \to N \to K \to 0$ is a SES of R-modules. Show that if M, K are flat then N is flat. Further show that the converse need not be true.

In this problem we use repeatedly the associativity of the tensor product. That is, if R, S are rings and $M \in \mathrm{Mod}_R, N \in \mathrm{Mod}_{R,S}, K \in \mathrm{Mod}_S$, then

$$(M \otimes_R N) \otimes_S K \cong M \otimes_R (N \otimes_S K)$$

naturally as S-modules. Indeed, if L is any S-module by the tensor-hom adjunction,

$$\hom((M \otimes_R N) \otimes_S K, L) \cong \hom(M \otimes_R N, \hom(K, L)) \cong \hom(M, \hom(N, \hom(K, L))).$$

Similarly,

$$\hom(M \otimes_R (N \otimes_S K), L) \cong \hom(M, \hom(N \otimes_S K, L)) \cong \hom(M, \hom(N, \hom(K, L))),$$

all naturally. So

$$\hom((M \otimes_R N) \otimes_S K, \bullet) \cong \hom(M \otimes_R (N \otimes_S K), \bullet)$$

naturally, and since the Yoneda embedding is free and faithful, and thus reflects isomorphisms, $(M \otimes_R N) \otimes_S K \cong M \otimes_R (N \otimes_S K)$.

(1) We want to show that $\bullet \otimes_R M|_R$ is exact (where $\bullet|_R$ denotes restriction of an A-module to an R-module). For any R-module N, we have (we omit explicitly writing $\bullet|_R$)

$$N \otimes_R M \cong N \otimes_R (A \otimes_A M) \cong (N \otimes_R A) \otimes_A M$$

naturally. That is $\bullet \otimes_R M \cong (\bullet \otimes_R A) \otimes_A M$. Since A is a flat R-module, $\bullet \otimes_R A$ is exact, and M is a flat A-module so tensoring with it over A is exact. Thus $\bullet \otimes_R M$ is naturally isomorphic to the composition of exact functors, and is thus exact, so M is flat.

(2) Let N be an A-module, then

$$N \otimes_A (A \otimes_R M) \cong (N \otimes_A A) \otimes_R M \cong N \otimes_R M.$$

Thus $\bullet \otimes_A (A \otimes_R M) \cong \bullet \otimes_R M$ which is exact, so $A \otimes_R M$ is flat.

(3) We know $S^{-1}M \cong (S^{-1}R) \otimes_R M$, so this follows from the previous point.

(4) If $\operatorname{Spec}(A) \to \operatorname{Spec}(R)$ is flat, then consider the open covering of $\operatorname{Spec}(R)$ by the basic open set $\operatorname{Spec}(R) = \operatorname{Spec}(R)_1$. Then

$$\operatorname{Spec}(A) \times_{\operatorname{Spec}(R)} \operatorname{Spec}(R) \cong \operatorname{Spec}(A),$$

and the map from the fiber product to $\operatorname{Spec}(R)$ just corresponds to the map $\operatorname{Spec}(A) \to \operatorname{Spec}(R)$, which is flat by assumption.

Conversely, we have flat maps $R_{f_i} \to R_{f_i} \otimes_R A \cong A_{f_i}$ for an open cover. Since the $\operatorname{Spec}(R)_{f_i}$ form an open cover, for every $\mathfrak{p} \in \operatorname{Spec}(R)$ there is an f_i such that $\mathfrak{p} \in \operatorname{Spec}(R)_{f_i}$, meaning $f_i \notin \mathfrak{p}$. In particular this means that $\mathfrak{p}$ does not intersect the localizing set $\{f_i^k\}_{k \geq 0}$ and so $\mathfrak{p}R_{f_i}$ is a prime ideal. Since A_{f_i} is a flat R_{f_i}-module, this means that $(A_{f_i})_{\mathfrak{p}R_{f_i}}$ is a flat $(R_{f_i})_{\mathfrak{p}R_{f_i}}$-module.

In general we know that if $S, T \subseteq R$ are multiplicative and $\overline{T} \subseteq S^{-1}R$ is the image of T in $S^{-1}R$ (which remains multiplicatively closed), then $\overline{T}^{-1}S^{-1}R \cong (S \cup T)^{-1}R$ naturally. This can be shown by appealing to the universal property of localization. Consider the canonical $\iota_0 \colon R \to (S \cup T)^{-1}R$ which maps $S \cup T$ to units. In particular it maps S to units and thus factors through $R \to S^{-1}R$, and since it maps T to units it must map $\overline{T}$ to units, and thus it factors through $S^{-1}R \to \overline{T}^{-1}S^{-1}R$. So we have a unique map $\overline{T}^{-1}S^{-1}R \to (S \cup T)^{-1}R$, and similarly we have a unique map in the opposite direction (unique in the property of factorizing through the relevant maps). By uniqueness, their compositions must be the identities. A similar result (proven the same way) holds for modules.

Since $\{f_i^k\}_k$ is contained in $R - \mathfrak{p}$, we then see that $(A_{f_i})_{\mathfrak{p}R_{f_i}} \cong ((R - \mathfrak{p}) \cup \{f_i\}^k)^{-1}A = A_{\mathfrak{p}}$. So $A_{\mathfrak{p}}$ is a flat $R_{\mathfrak{p}}$ module for every prime $\mathfrak{p}$. Thus by the locality of flatness, A is a flat R-module as desired.

(5) Let L be an arbitrary R-module, then the Tor functor gives us an exact sequence

$$\operatorname{Tor}_1^R(M, L) \to \operatorname{Tor}_1^R(N, L) \to \operatorname{Tor}_1^R(K, L).$$

Since M, K are flat we know $\operatorname{Tor}_1^R(M, L) = \operatorname{Tor}_1^R(K, L) = 0$, and so by exactness $\operatorname{Tor}_1^R(N, L) = 0$. This means that N is flat, as desired.

Now consider the flat $\mathbb{Z}$-module, $\mathbb{Z}$ (it is free and thus flat). We have an SES

$$0 \to (2) \to \mathbb{Z} \to \mathbb{Z}/2 \to 0.$$

Here both (2) and $\mathbb{Z}$ are free, and thus flat, but $\mathbb{Z}/2$ is not flat. This is because multiplication by 2 in $\mathbb{Z}$ is injective $\mathbb{Z} \to \mathbb{Z}$, but tensoring with $\mathbb{Z}/2$ gives the map $\mathbb{Z}/2 \to \mathbb{Z}/2$ which maps $x \mapsto 2x = 0$ and is thus not injective. So tensoring with $\mathbb{Z}/2$ doesn't preserve injections, and thus it is not flat. So N being flat is not enough to deduce that K is.

And now consider $R = \mathbb{C}[x, y]$, which is a flat R-module, and $I = (x, y) \leq R$. So we have an SES

$$0 \to I \to R \to R/I \to 0.$$

But we claim that I is not a flat R-module. Consider the ideal $J = (y) \leq R$ and the injection $R/J \to R/J$ given by multiplication by x. This is an injection because $x \cdot f(x, y) \in J$ iff $f(x, y) \in J$ since J is prime.

Now recall that $R/J \otimes_R I \cong I/(IJ)$ by $(r + J) \otimes a \mapsto ra + IJ$. Under this identification, multiplication by x is tensored to give a map $I/(IJ) \to I/(IJ)$ given by $f + IJ \mapsto xf + IJ$. But this is not injective: $IJ = (x, y)(x) = (x^2, xy)$ and so $y \notin IJ$ is mapped to $xy \in IJ$. So tensoring with I does not preserve injections, and thus I is not flat.

6.2 Problem

Let M be an R-module. We will show that M is flat iff for all $a \in R^m$ and $x \in M^m$ ($m \geq 0$) such that $a^\top x = 0$, there exists some $n \geq 0$, a matrix $B \in \mathrm{Mat}_{m\times n}(R)$, and a vector $y \in M^n$ such that

$$a^\top B = 0, \qquad By = x.$$

(1) Assume M is flat, and let $a \in R^m, x \in M^m$ be such that $a^\top x = 0$.

(i) Consider the sequence $0 \to K \to R^m \to R$, where $R^m \to R$ is given by $b \mapsto a^\top b$, and K is the kernel of that map. Show that $K \otimes_R M \to M^m \to M$ is exact, where the map $M^m \to M$ is given by $t \mapsto a^\top t$.

(ii) Conclude that $x = \sum_{j=1}^n \beta_j y_j$ for some $n \geq 0$, and $\sum_{j=1}^n \beta_j \otimes y_j \in K \otimes_R M$. Let $B_{ij} = (\beta_j)_i$, and show that B and y satisfy the conclusion of the claim.

(2) In the other direction, let $I \leq R$ be an ideal and show that $I \otimes_R M \to IM$ is an isomorphism and conclude the result.

(1) Let us denote by $\varphi\colon R^m \to R$ the map given: $\varphi(b) = a^\top b$. So we have an exact sequence

$$0 \to K \to R^m \xrightarrow{\varphi} R.$$

Since M is flat, tensoring preserves exact sequences and thus the following is exact

$$0 \to K \otimes_R M \to R^m \otimes_R M \xrightarrow{\varphi\otimes M} R \otimes_R M.$$

We know $R^m \otimes_R M \cong M^m$ via $(b_i) \otimes m \mapsto (b_i m)$, and $R \otimes_R M \cong M$ via $b \otimes m \mapsto bm$. Thus this induces an exact sequence

$$0 \to K \otimes_R M \xrightarrow{\psi} M^m \xrightarrow{\varphi'} M,$$

where the first map is $(k_i) \otimes m \mapsto (k_i m)$, and the second is given by the composition

$$M^m \cong R^m \otimes_R M \xrightarrow{\varphi\otimes M} R \otimes_R M \cong M.$$

Let $e_i \in R^m$ be the standard basis vectors, so this is given by mapping a vector $(m_i) \in M^m$ through

$$(m_i) = \sum_i e_i m_i \mapsto \sum_i (e_i \otimes m_i) \xmapsto{\varphi\otimes M} \sum_i (a^\top e_i) \otimes m_i \mapsto \sum_i (a^\top e_i) m_i.$$

Since $a^\top e_i = a_i$ this is just the map

$$(m_i) \mapsto \sum_i a_i m_i = a^\top m,$$

as desired.

Since the sequence is exact, we have that $\mathrm{im}\psi = \ker \varphi'$. And by assumption $a^\top x = 0$ so $x \in \mathrm{im}\psi$ so there are $\beta_j \otimes y_j \in K \otimes_R M$, for $j = 1, \ldots, n$ for some n, such that

$$\psi'\left(\sum_j \beta_j \otimes y_j\right) = \sum_j \beta_j y_j = x.$$

Now $\beta_j \in K \subseteq R^m$ so define $B \in \mathrm{Mat}_{n\times m}(R)$ by $B_{ij} = (\beta_j)_i$. Then

$$By = \sum_j \mathrm{Col}_j(B) y_j = \sum_j \beta_j y_j = x.$$

And since for any $z \in M^n$ we have that $Bz \in \mathrm{im}\psi$ (since $\beta_j \in K$), and thus $a^\top Bz = 0$. This implies that $a^\top B = 0$ (take $z = e_i$, this shows that the ith column of $a^\top B$ is zero), as desired.

(2) The map $I \otimes_R M \to IM$ is always a surjection, so we need only show it is an injection. Suppose that $\sum_{i=1}^m a_i \otimes m_i \in I \otimes_R M$ is in the kernel, so $\sum_i a_i m_i = 0$. Letting $a = (a_i) \in R^m$ and $x = (m_i) \in M^m$ by assumption there is a matrix $B \in \mathrm{Mat}_{m\times n}(R)$ and a vector $y \in R^n$ such that $a^\top B = 0$ and $By = x$.

Now let us define the following: for $a \in R^m$ and $x \in M^m$, define

$$a^\top \otimes x = \sum_{i=1}^m a_i \otimes x_i.$$

Now we claim that for any $A \in \mathrm{Mat}_{m\times n}(R)$ and $y \in M^n$,

$$a^\top \otimes (Ay) = (A^\top a) \otimes y = (a^\top A)^\top \otimes y.$$

Indeed,

$$a^\top \otimes (Ay) = \sum_{i=1}^m a_i \otimes (Ay)_i = \sum_{i=1}^m a_i \otimes \sum_{j=1}^n A_{ij} y_j = \sum_{i,j} A_{ij} a_i \otimes y_j.$$

and

$$(A^\top a) \otimes y = \sum_{j=1}^n (A^\top a)_j \otimes y_j = \sum_{j,i} A^\top_{ji} a_i \otimes y_j = \sum_{j,i} A_{ij} a_i \otimes y_j,$$

as desired.

Now in our case, we have

$$\sum_{i=1}^m a_i \otimes m_i = a^\top \otimes x = a^\top \otimes (By) = (a^\top B)^\top \otimes y = 0 \otimes y = 0,$$

as desired. So the map is indeed an injection, and thus M is flat.

6.3 Problem

Fix a ring R and a fg R-module M.

(1) Assume R is a field, show that M is flat.

(2) Assume R is a local ring with maximal ideal $\mathfrak{m}$. Show that M is flat iff it is free.

(3) Assume R is a PID, show that M is flat iff it is torsion-free.

(1) If R is a field, all modules over it are free (linear algebra) and thus flat.

(2) If M is free it is flat, so assume that M is flat and we will show it is free.

Nakayama's lemma says that $x_1, \dots, x_n \in M$ generate it iff their projections generate $M/\mathfrak{m}M$ as an $R/\mathfrak{m}$-module. So choose a basis for $M/\mathfrak{m}M$ (since $R/\mathfrak{m}$ is a field), and let $x_1, \dots, x_n \in M$ be their preimages. So by Nakayama, $x_1, \dots, x_n \in M$ generate M and their projections form a basis.

We claim that $x_1, \dots, x_n$ are linearly independent. Suppose not, so there is an $a \in R^n$ such that $a^\top x = 0$. By the equational criterion proven above, there is a $B \in \mathrm{Mat}_{m\times n}(R)$ and $y \in M^m$ such that $By = x$ and $a^\top B = 0$.

We induct on n to show that if $a^\top x = 0$ for a vector $x = (x_1, \dots, x_n)$ which are linearly independent in $M/\mathfrak{m}M$, then $a = 0$. For $n = 1$, if x_1 is a basis for $M/\mathfrak{m}M$ then $x_1 \notin \mathfrak{m}M$ and if $By = x_1$ then $y = (y_1, \dots, y_m)$ and $B = (b_1, \dots, b_n)^\top$ so $x_1 = \sum_i b_i y_i$ and $ab_i = 0$ for all i. Now $x_1 \notin \mathfrak{m}M$ and thus we must have that there is a b_i which is not in $\mathfrak{m}$. Since R is local, this means b_i is a unit, and thus $a = 0$.

Now for $n > 1$ we have $\sum_i a_i x_i = 0$. Let $\overline{x}$ denote the projection of $x \in M$ to $M/\mathfrak{m}M$ and $\overline{a}$ project R to $R/\mathfrak{m}$. So we have

$$\sum_i \overline{a}_i \overline{x}_i = 0 \Longrightarrow \overline{a}_i = 0 \text{ for all } i.$$

Meaning $a_i \in \mathfrak{m}$ for all i. And we have

$$x_i = \sum_j b_{ij} y_j, \qquad \sum_i a_i b_{ij} = 0 \text{ for all } j.$$

Now since $x_n \notin \mathfrak{m}M$, we see that we must have some $b_{nj} \notin \mathfrak{m}$ and thus b_{nj} is a unit for the same reason as before. Thus for this j,

$$\sum_i a_i b_{ij} = 0 \Longrightarrow a_n = -\sum_{i<n} a_i b_{ij}/b_{nj}.$$

Now

$$0 = \sum_i a_i x_i = \sum_{i<n} a_i x_i - \sum_{i<n} a_i (b_{ij}/b_{nj}) x_n = \sum_{i<n} a_i (x_i - (b_{ij}/b_{nj}) x_n).$$

Since the x_i are linearly independent in $\mathfrak{m}M$, $x_i - (b_{ij}/b_{nj})x_n$ are linearly independent for $i = 1, \dots, n-1$. By induction, we have $a_i = 0$ for $i < n$. Then we have $a_n x_n = 0$ and by the base case, $a_n = 0$ as well.

So $x_1, \dots, x_n$ are linearly independent over M and thus form a bases and M is free as desired.

(3) If M is flat then it is torsion-free. Indeed, if $x \in M$ and $a \in R$ a non-zero divisor such that $ax = 0$ then by the equational criterion there are $b_1, \dots, b_n \in R$ such that $ab_i = 0$ for all i and $y_1, \dots, y_n \in M$ such that $\sum_i b_i y_i = x$. Since a is a non-zero divisor, $b_i = 0$ for all i and thus $x = 0$.

Conversely if M is torsion-free, then let $I = (a)$ be an ideal of R. Then $IM = (a)M = aM = \{am \mid m \in M\}$, and define $aM \to (a) \otimes_R M$ by $bm \mapsto b \otimes m$. This is well-defined: if $bm = bn$ then $b(m - n) = 0$ and since M is torsion-free and R is a domain, $m = n$. And this is an inverse to $(a) \otimes_R M \to aM$, $ab \otimes m \mapsto abm$. Indeed, the composition maps $ab \otimes m \mapsto abm \mapsto ab \otimes m$, and composition in the other direction maps $abm \mapsto ab \otimes m \mapsto abm$. Thus $(a) \otimes_R M \to aM$ is an isomorphism for all ideals $I = (a)$, and thus M is flat.

Note that these proofs work for modules which aren't finitely-generated.

6.4 Problem

For each ring R and R-module M, determine if M is flat.

(1) $R = \mathbb{Z}$ and $M = \mathbb{Q}$,

(2) $R = \mathbb{C}[x, y]$ and $M = (x, y)$,

(3) $R = \mathbb{C}[x, y]/(y^2 - x^3 - x)$ and $M = (x, y)$.

(1) M is the localization of $\mathbb{Z}$ at $\mathbb{Z} - 0$, and since $\mathbb{Z}$ is a flat $\mathbb{Z}$-module and localization preserves flatness, M is flat.

(2) This was the example I gave in the first question of a module which is not flat.

(3) We first note that $f = y^2 - x^3 - x$ is irreducible. If it factored, it would have to factor as $f = (y - g(x))(y + g(x))$, but that would imply $x^3 + x$ is a square $g(x)^2$ which it is not. Thus R is an integral domain, and since $M \leq R$ is an ideal it is torsion-free.

Now consider all the prime ideals containing f, these are $\mathfrak{m} = (x - a, y - b)$ such that $f(a, b) = 0$ i.e. $b^2 = a^3 + a$. Thus the maximal ideals of R are quotients of these $\mathfrak{m}$, which we identify with $\mathfrak{m}$. We will show that $M_\mathfrak{m}$ is a flat $R_\mathfrak{m}$-module, and then by locality M is flat.

If $\mathfrak{m} \neq (x, y)$ then either x or y is not in $\mathfrak{m}$ and thus becomes a unit in $R_\mathfrak{m}$, and then $M_\mathfrak{m} = (x, y)_\mathfrak{m} = R_\mathfrak{m}$ is free and thus flat. Otherwise $\mathfrak{m} = (x, y)$, and then $1 + x^2 \notin \mathfrak{m}$ so $1 + x^2$ is a unit in $R_\mathfrak{m}$. Then $y^2 = x(1 + x^2)$ implies $x = (1 + x^2)^{-1}y^2$, so $M_\mathfrak{m} = (x, y)_\mathfrak{m} = (y)_\mathfrak{m}$ is a principal ideal in $R_\mathfrak{m}$. Since R is an integral domain, localization remains an integral domain, and principal ideals in an integral domain are free. Thus $M_\mathfrak{m}$ is a free, and thus flat, $R_\mathfrak{m}$-module.

So $M_\mathfrak{m}$ is flat for all maximal $\mathfrak{m} \leq R$, and thus M is flat.

6.5 Problem

Let R be a ring and M a flat R-module. M is called **faithfully flat** if tensoring with it reflects exactness: $0 \to N \to K \to L \to 0$ is exact iff $0 \to N \otimes M \to K \otimes M \to L \otimes M \to 0$ is exact.

(1) Show that the following are equivalent for a flat module M.

- **(i)** M is faithfully flat,
- **(ii)** $N \otimes M = 0$ iff $N = 0$ for all R-modules N,
- **(iii)** For all $\mathfrak{p} \in \operatorname{Spec}(R)$, $k(p) \otimes M \neq 0$,
- **(iv)** For any maximal $\mathfrak{m} \leq R$, $M/\mathfrak{m}M \neq 0$.

(2) Let A be a flat R-algebra. Show that the following are equivalent:

- **(i)** A is faithfully-flat,
- **(ii)** $\operatorname{Spec}(A) \to \operatorname{Spec}(R)$ is surjective,
- **(iii)** All closed points of $\operatorname{Spec}(R)$ are in the image of $\operatorname{Spec}(A) \to \operatorname{Spec}(R)$.

(3) Let $\varphi\colon R \to A$ be a morphism of local rings, i.e. R and A are local with maximal ideals $\mathfrak{m}, \mathfrak{n}$ and $\varphi^{-1}\mathfrak{n} = \mathfrak{m}$. Assume A is flat, show that it is faithfully-flat.

(1) We show the equivalence of the first two points by a slight generalization: an additive exact functor F is faithfully exact (i.e. reflects exactness) if $F(M) = 0$ only when $M = 0$. If F is faithfully flat and $F(M) = 0$, then $F(0) = 0 \to F(M) \to 0 = F(0)$ is exact, and thus by faithfullness $0 \to M \to 0$ is exact and thus $M = 0$. Conversely if $F(M) = 0$ implies $M = 0$, then suppose $F(M) \xrightarrow{Ff} F(N) \xrightarrow{Fg} F(K) \to 0$ is exact. If $F(g \circ f) = 0$ then since F is exact it commutes with images (as they can be written as kernels of cokernels, and F commutes with those), so $0 = \operatorname{im}F(g \circ f) = F(\operatorname{im} g \circ f)$, and thus $\operatorname{im} g \circ f = 0$ so $g \circ f = 0$. Thus $\operatorname{im} f \subseteq \ker g$. Furthermore exact functors commute with quotients (for a similar reason as before), so

$$F(\ker g/\operatorname{im} f) \cong \frac{F \ker g}{F \operatorname{im} f} \cong \frac{\ker Fg}{\operatorname{im} Ff} = 0,$$

where the final equality is due to exactness of the sequence. Thus $\ker g/\operatorname{im} f = 0$ and thus $\operatorname{im} f = \ker g$, so F reflects exactness.

Since $k(\mathfrak{p}) \neq 0$, if M is faithfully flat then $k(\mathfrak{p}) \otimes M \neq 0$. Conversely suppose that $k(\mathfrak{p}) \otimes M \neq 0$ for all prime $\mathfrak{p}$, then if $N \otimes M = 0$, we have

$$0 = R_{\mathfrak{p}} \otimes N \otimes M \cong N_{\mathfrak{p}} \otimes M \cong (N_{\mathfrak{p}} \otimes_{k(\mathfrak{p})} k(\mathfrak{p})) \otimes M \cong N_{\mathfrak{p}} \otimes_{k(\mathfrak{p})} (k(\mathfrak{p}) \otimes M).$$

But over a field K, $V \otimes_K U = 0$ iff $V = 0$ or $U = 0$ (considering dimensions), so this implies $N_{\mathfrak{p}} = 0$. This is true for all prime $\mathfrak{p}$, and thus $N = 0$.

Note the above argument works when replacing prime ideals with maximal ideals. And $k(\mathfrak{m}) \cong R/\mathfrak{m}$, so $k(\mathfrak{p}) \otimes M \cong M/\mathfrak{m}M$. Thus M is faithfully flat iff $M/\mathfrak{m}M \neq 0$ for all maximal $\mathfrak{m}$.

(2) Let $\varphi\colon R \to A$ be the algebra morphism.

If A is faithfully flat, then let $\mathfrak{p} \in \operatorname{Spec}(R)$. Then $k(\mathfrak{p}) \otimes_R A \neq 0$, so

$$0 \neq k(\mathfrak{p}) \otimes_R A = \frac{R_{\mathfrak{p}}}{\mathfrak{p}R_{\mathfrak{p}}} \otimes_R A \cong \frac{R}{\mathfrak{p}} \otimes_R R_{\mathfrak{p}} \otimes_R A \cong \frac{R}{\mathfrak{p}} \otimes_R A_{\mathfrak{p}} \cong \frac{A_{\mathfrak{p}}}{\mathfrak{p}A_{\mathfrak{p}}}.$$

So $A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$ is a nonzero R-algebra, so $\mathfrak{p}A_{\mathfrak{p}} \neq A_{\mathfrak{p}}$. Recall that $A_{\mathfrak{p}}$ is isomorphic to the localization of A by $\varphi(R - \mathfrak{p})$, and $\mathfrak{p}A_{\mathfrak{p}}$ is equal to $\varphi(\mathfrak{p})A_{\mathfrak{p}}$. Since $\mathfrak{p}A_{\mathfrak{p}}$ is a proper ideal of $A_{\mathfrak{p}}$, it is contained in a prime ideal of $A_{\mathfrak{p}}$. Such an ideal is of the form $\mathfrak{q}A_{\mathfrak{p}}$ for $\mathfrak{q} \in \operatorname{Spec}(A)$ and $\mathfrak{q} \cap \varphi(R - \mathfrak{p}) = \emptyset$.

Thus we have that $\varphi^{-1}\mathfrak{q} \leq \mathfrak{p}$ since if $\varphi(x) \in \mathfrak{q}$ then $\varphi(x) \notin \varphi(R - \mathfrak{p})$ and so $x \in \mathfrak{p}$. But on the other hand, $\varphi(\mathfrak{p})A_{\mathfrak{p}} \leq \mathfrak{q}A_{\mathfrak{p}}$ so $\varphi(\mathfrak{p}) \leq \mathfrak{q}$ and thus $\mathfrak{p} \leq \varphi^{-1}\mathfrak{q}$. Thus we have that $\mathfrak{p} = \varphi^*(\mathfrak{q})$ so φ^* is surjective.

And if $\operatorname{Spec}(A) \to \operatorname{Spec}(R)$ is surjective obviously its image contains all the closed points.

And if φ^*'s image contains all the closed points, i.e. maximal ideals $\mathfrak{m} \leq R$ then A is faithfully flat. Indeed, let $\mathfrak{m} \in \operatorname{Spec}(R)$ so $\mathfrak{m} = \varphi^{-1}\mathfrak{q}$ for some $\mathfrak{q} \in \operatorname{Spec}(A)$. Then $\mathfrak{m}A = \varphi^{-1}(\mathfrak{q})A$ is a subset of $\mathfrak{q}$, since $\varphi^{-1}(\mathfrak{q})A$ is the ideal generated by $\varphi(\varphi^{-1}\mathfrak{q})$ by definition, which is a subset of $(\mathfrak{q}) = \mathfrak{q}$. Since $\mathfrak{q} \neq A$, we have $\mathfrak{m}A \neq A$ and thus $A/\mathfrak{m}A \neq 0$ for all maximal $\mathfrak{m} \in \operatorname{Spec}(R)$, so A is faithfully flat.

(3) If φ is a morphism of local rings then by definition φ^*'s image contains all maximal ideals, i.e. closed points, and thus if A is flat it is faithfully flat.

6.6 Problem

Fix a morphism of rings $\varphi\colon B \to A$, and denote $X = \operatorname{Spec}(A), Y = \operatorname{Spec}(B)$.

(1) Assume φ is flat, and let $I, J \leq B$. Prove that $(I \cap J)A = IA \cap JA$.

(2) Find a counterexample to the above point when φ is not flat.

(3) Assume φ is flat. Prove the **going-down property**: for any $\mathfrak{q}_1, \mathfrak{q}_2 \in Y$ such that $\mathfrak{q}_1 \in \overline{\{\mathfrak{q}_2\}}$, and $\mathfrak{p}_1 \in X$ such that $\varphi^*(\mathfrak{p}_1) = \mathfrak{q}_1$, then there exists some $\mathfrak{p}_2 \in X$ such that $\mathfrak{p}_1 \in \overline{\{\mathfrak{p}_2\}}$ and $\varphi^*(\mathfrak{p}_2) = \mathfrak{q}_2$.

(1) By definition, we have

$$(I \cap J)A = (\varphi(I \cap J)) = (\varphi(I) \cap \varphi(J)) \subseteq (\varphi(I)) \cap (\varphi(J)) = IA \cap JA.$$

So one inclusion is true in general.

If A is flat, consider the exact sequence

$$0 \to I \cap J \to B \to (B/I) \oplus (B/J).$$

Since A is flat, tensoring is exact and thus the following sequence is exact

$$0 \to (I \cap J) \otimes A \to A \to (A/IA) \oplus (A/JA),$$

where the first map is $b \otimes a \mapsto ba$, then second is the obvious sum of projections (we use that $A \otimes B \cong A$, and $A \otimes (B/I) \cong A/(IA)$). Since A is flat, $(I \cap J) \otimes A \cong (I \cap J)A$, and thus

$$0 \to (I \cap J)A \to A \to (A/IA) \oplus (A/JA)$$

is exact, where the first map is the inclusion map.

Thus $(I \cap J)A$ is equal to the kernel of $A \to (A/IA) \oplus (A/JA)$, which is just $IA \cap JA$, as desired.

(2) Consider $B = \mathbb{Z}[x, y]$ and $A = \mathbb{Z}[x, y, z]/(zx - zy)$, and $I = (x), J = (y)$. Then notice that $zx \in (x)A$ so $zx = zy \in (x)A \cap (y)A$. But $I \cap J = (xy)$, and $zx \notin (xy)A$. This is because we can consider $A/(xy) = \mathbb{Z}[x, y, z]/(zx - zy, xy)$, and quotienting over a larger ideal,

$$\frac{\mathbb{Z}[x, y, z]}{(x - y, xy)} \cong \frac{\mathbb{Z}[x, z]}{(x^2)} \cong \frac{\mathbb{Z}[x]}{(x^2)}[z].$$

And zx is nonzero here clearly since x is nonzero in $\mathbb{Z}[x]/(x^2)$, thus zx is nonzero in $A/(xy)$, and thus $zx \notin (xy)A$.

(3) Since $B \to A$ is flat, so is $B_{\mathfrak{q}_1} \to A_{\mathfrak{q}_1}$ (flatness is local). This is actually faithfully flat, since $\varphi^{-1}(\mathfrak{p}_1 A_{\mathfrak{q}_1}) = \mathfrak{q}_1 B_{\mathfrak{q}_1}$ is the unique maximal ideal of $B_{\mathfrak{q}_1}$, so the spectrum map has all of the closed points in its image, and thus is faithfully flat by the previous question. Thus the map $\mathrm{Spec}(A_{\mathfrak{q}_1}) \to \mathrm{Spec}(B_{\mathfrak{q}_1})$ is surjective. The spectrum of $\mathrm{Spec}(A_{\mathfrak{p}})$ corresponds to all the prime ideals of A which are contained in $\mathfrak{p}$ (all those $\mathfrak{q} \in \mathrm{Spec}(A)$ such that $\mathfrak{p} \in \overline{\{\mathfrak{q}\}}$). So let $\varphi^*(\mathfrak{p}_2 A_{\mathfrak{q}_1}) = \mathfrak{q}_2 B_{\mathfrak{q}_2}$, then it follows $\varphi^*(\mathfrak{p}_2) = \mathfrak{q}_2$, and $\mathfrak{p}_1 \in \overline{\{\mathfrak{p}_2\}}$.

6.7 Problem

Fix a ring R and two R-modules M, N.

(1) Show that N has a free resolution: free R-modules $N_0, N_1, \ldots$ such that there is a long exact sequence

$$\mathscr{R} \to N \colon \cdots \to N_1 \to N_0 \to N \to 0.$$

(2) Show that tensoring with M forms a chain complex $\mathscr{R} \otimes M \to N \otimes M \to 0$. Define $\mathrm{Tor}^n(M, N)$ to be the nth homology module of $R \otimes M$.

(3) Show that $\mathrm{Tor}_0^R(M, N) \cong M \otimes N$.

(4) Given another module N' and a free resolution $\mathscr{R}' \to N' \to 0$, show that a morphism $N \to N'$ lifts to a morphism of resolutions, i.e. a commutative diagram:

$$\begin{tikzcd}
\cdots \arrow[r] & N_1 \arrow[r] \arrow[d] & N_0 \arrow[r] \arrow[d] & N \\
\cdots \arrow[r] & N_1' \arrow[r] & N_0' \arrow[r] & N'
\end{tikzcd}$$

(5) Show that such a lift induces a morphism of chain complexes.

(6) Given two distinct lifts, show that the two induced maps are homotopic (two maps between chain complexes $f, g \colon C \to D$ are **homotopic** if there are maps $s_i \colon C_i \to D_{i+1}$ such that $f - g = ds + sd$).

(7) Show that a lift induces a map $\mathrm{Tor}_i^R(M, N) \to \mathrm{Tor}_i^R(M, N')$.

(8) Show that homotopic maps induce the same map between torsion modules.

(9) Let $\mathscr{R}, \mathscr{R}'$ be two free resolutions of N, show that they define isomorphic torsion modules. Thus $\mathrm{Tor}^R(\bullet, M)$ is well-defined.

(10) Show that $\mathrm{Tor}_i^R(\bullet, \bullet)$ is a bifunctor $\mathrm{Mod}_R \times \mathrm{Mod}_R \to \mathrm{Mod}_R$.

(11) Show that $\mathrm{Tor}^R(M, \bullet)$ maps a short exact sequence to long exact sequences.

(12) Assume that N is free, and let $i > 0$. Show that $\mathrm{Tor}_i^R(N, M) = 0$.

(13) Show that M is flat iff $\mathrm{Tor}_i^R(M, \bullet) = 0$ for all $i > 0$ iff $\mathrm{Tor}_1^R(M, \bullet) = 0$.

(1) For every R-module N there is a surjective map $N_0 \xrightarrow{d_0} N \to 0$ where N_0 is free (e.g. let $N_0 = R^{\oplus N}$ and map $e_n \mapsto n$ for $n \in N$). Now suppose we have defined $d_n \colon N_n \to N_{n-1}$. Consider then $\ker d_n \subseteq N_n$ and let $d_{n+1} \colon N_{n+1} \to \ker d_n$ be surjective where N_{n+1} is free, as before. Then $d_{n+1} \colon N_{n+1} \to N_n$ has image $\ker d_n$, and thus the chain is exact, and we have a free resolution.

(2) Additive functors, like tensoring with M, map chain complexes to chain complexes. This is because if $g \circ f = 0$ then

$$F(g) \circ F(f) = F(g \circ f) = F(0) = 0.$$

(3) Let F be right-exact, like $\bullet \otimes M$, then if $N_1 \xrightarrow{d_1} N_0 \xrightarrow{d_0} N \to 0$ is exact then so is $F(N_1) \to F(N_0) \to F(N) \to 0$. Thus $\mathrm{im} F(d_1) = \ker F(d_0)$. And we want to show that the homology module of $F(N_1) \to F(N_0) \to 0$ is isomorphic to $F(N)$.

The homology module is $F(N_0)/\mathrm{im} F(d_1) = F(N_0)/\ker F(d_0) \cong F(N)$ where the last isomorphism is due to the isomorphism theorem and the surjectivity of $F(d_0) \colon F(N_0) \to F(N)$.

(4) Suppose the free resolution $\mathscr{R}$ has differentials $d_\bullet$ and $\mathscr{R}'$ has differentials $d'_\bullet$. Given $f \colon N \to N'$ we inductively define $f_i \colon N_i \to N'_i$ such that $d'_i \circ f_i = f_{i-1} \circ d_i$ (where $f_{-1} = f$).

For $i = 0$ we need $d'_0 \circ f_0 = f \circ d_0$. It is sufficient to find images for $x \in N_0$ where the xs are from a basis for N_0 and extend by linearity, as it is free and the condition ($d' \circ f = f \circ d$) is linear. By assumption $d'_0 \colon N'_0 \to N'$ is surjective, so certainly for every x there is an $f_0(x)$ such that $d'_0(f_0(x)) = f \circ d_0(x)$, as desired.

Now assuming we have constructed f_i we construct f_{i+1}, again we need only find a value of $f_{i+1}(x)$ for an individual x and use this to define the values on a basis and extend by linearity. We need to find a $y \in N_{i+1}$ such that

$$d'_{i+1}(y) = f_i \circ d_{i+1}(x).$$

That is, we want to show $f_i \circ d_{i+1}(x) \in \mathrm{im} d'_{i+1}$. By exactness this is equal to the kernel of d'_i so it is sufficient to show $d'_i \circ f_i \circ d_{i+1} = 0$. Inductively we have

$$d'_i \circ f_i = f_{i-1} \circ d_i \Longrightarrow d'_i \circ f_i \circ d_{i+1} = f_{i-1} \circ d_i \circ d_{i+1} = 0$$

as desired. So such a y exists, and we can set $f_{i+1}(x) = y$.

(5) Tensoring is functorial, it maps commutative diagrams to commutative diagrams.

(6) If F is an additive functor, then homotopic maps $\mathscr{R} \to \mathscr{R}'$ are mapped to homotopic maps $F(\mathscr{R}) \to F(\mathscr{R}')$. This is because

$$F(f) - F(g) = F(f - g) = F(ds + sd) = F(d)F(s) + F(s)F(d),$$

so $F(s)$ forms the homotopy. So it is sufficient to show that any two lifts of $f \colon N \to N'$ are homotopic, then the induced maps by $\bullet \otimes M$ are homotopic.

If we have two lifts f_i, g_i then $f_{-1} = g_{-1} = f$ and so choosing $s_{-1} = 0$ gives us our base. Now suppose that inductively we have $f_i - g_i = d'_{i+1}s_i + s_{i-1}d_i$. So we need to define $s_{i+1}\colon N_{i+1} \to N'_{i+2}$ such that

$$f_{i+1} - g_{i+1} = d'_{i+2} \circ s_{i+1} + s_i \circ d_{i+1},$$

equivalently

$$d'_{i+2} \circ s_{i+1} = f_{i+1} - g_{i+1} - s_i \circ d_{i+1}.$$

As before it is sufficient to find a value $y = s_{i+1}(x)$ for any $x \in N_{i+1}$ by freeness. So we want to show that $\mathrm{im}(f_{i+1} - g_{i+1} + s_i \circ d_{i+1}) \in \mathrm{im} d'_{i+2} = \ker d'_{i+1}$. Thus we compute

$$\begin{aligned} d'_{i+1}(f_{i+1} - g_{i+1} - s_i \circ d_{i+1}) &= (f_i - g_i) \circ d_{i+1} - d'_{i+1} \circ s_i \circ d_{i+1} \\ &= (d'_{i+1} \circ s_i + s_{i-1} \circ d_i) \circ d_{i+1} - d'_{i+1} \circ s_i \circ d_{i+1} \\ &= d'_{i+1}s_i d_{i+1} - d'_{i+1}s_i d_{i+1} = 0. \end{aligned}$$

As desired.

(7) We just need to show that if we have a map of chain complexes $f\colon C \to D$ then it induces a map between their homology modules. Indeed, since $d'_{i+1} \circ f_{i+1} = f_i \circ d_{i+1}$, we see that f_i maps boundaries to boundaries. And if $d_i x = 0$ then $d'_i \circ f_i(x) = f_{i-1} \circ d_i(x) = 0$, so $f_i(x)$ is a cycle meaning f_i maps cycles to cycles.

Thus we can restrict $f_i\colon C_i \to D_i$ to $f_i\colon Z_i(C) \to Z_i(D)$ and then since $f_i(B_i(C)) \subseteq B_i(D)$, we can quotient to give a map $H_i(f)\colon H_i(f) \to H_i(D)$. Note that this is clearly functorial: the identity restricts to the identity on cycles, and quotienting the identity gives the identity. And $H_i(f \circ g) = H_i(f) \circ H_i(g)$ since both are just the map $[x] \mapsto [f(g(x))]$ where $[x]$ is the coset of the cycle x. Furthermore since quotienting and restricting are additive, H_i is an additive functor.

In particular if we have a map between resolutions $f\colon (\mathscr{R} \to N) \to (\mathscr{R}' \to N')$, this lifts to a map of chain complexes $f \otimes M\colon (\mathscr{R} \otimes M) \to (R' \otimes M)$ which then quotient to give maps between the homology modules, which are just the torsion modules.

(8) It is sufficient to show that homotopic maps between chain complexes $f, g\colon C \to D$ induce equal maps on the homology modules. Since homology is additive, it is sufficient to show that $H_n(f - g) = 0$ in order to show that $H_n(f) = H_n(g)$. That is, we need to show that for any chain homotopy s, $H_n(sd + d's) = 0$.

Notice that $d's$ maps into boundaries, and thus quotients to zero. So we need only show that $H_n(sd) = 0$. But sd is just the zero map on cycles: if $a \in Z_i(C)$ then $da = 0$ by definition so $sda = 0$, so we get the desired result.

So in particular two lifts of the same map $f\colon N \to N'$, since they are homotopic and thus lift to homotopic maps $\mathscr{R} \otimes M \to \mathscr{R}' \otimes M$, induce the same maps on torsion modules.

(9) If $\mathscr{R}, \mathscr{R}'$ are two free resolutions of N, by above there is a unique (up to homotopy) extension of $\mathrm{id}\colon N \to N$ to $F\colon \mathscr{R} \to \mathscr{R}'$, and by the same reason to $G\colon \mathscr{R}' \to \mathscr{R}$. Now, $F \circ G\colon \mathscr{R}' \to \mathscr{R}'$ is also an extension of the identity, but $\mathrm{id}\colon \mathscr{R}' \to \mathscr{R}'$ is already an extension of the identity so they are chain homotopic. Thus $\mathrm{Tor}(M, F \circ G) = \mathrm{Tor}(M, \mathrm{id}) = \mathrm{id}$ and similarly $\mathrm{Tor}(M, G \circ F) = \mathrm{id}$. We show in the next point that the Tor is functorial (when considering any free resolution), so $\mathrm{Tor}(M, F), \mathrm{Tor}(M, G)$ are isomorphisms between $\mathrm{Tor}(M, N)_{\mathscr{R}}, \mathrm{Tor}(M, N)_{\mathscr{R}'}$.

(10) If we have a map $f\colon M \to M'$ then for any free resolution $\mathscr{R} \to N$ this induces a map of chain complexes $\mathscr{R} \otimes M \to \mathscr{R} \otimes M'$ whose ith component is $\mathrm{id}_{N_i} \otimes f\colon N_i \otimes M \to N_i \otimes M'$. This is a morphism of chain complexes since we consider the following square

$$\begin{tikzcd} N_i \otimes M \arrow[r, "d_i \otimes \mathrm{id}"'] \arrow[d, "\mathrm{id} \otimes f"'] & N_{i-1} \otimes M \arrow[d, "\mathrm{id} \otimes f"'] \\ N_i \otimes M' \arrow[r, "d_i \otimes \mathrm{id}"] & N_{i-1} \otimes M' \end{tikzcd}$$

we want to show it commutes. This is due to functorality of the tensor:

$$(\mathrm{id} \otimes f) \circ (d_i \otimes \mathrm{id}) = d_i \otimes f = (d_i \otimes \mathrm{id}) \circ (\mathrm{id} \otimes f).$$

Thus it quotients to give a map $\mathrm{id} \otimes f \colon \mathrm{Tor}^R(M, N) \to \mathrm{Tor}^R(M', N)$. That is,

$$\mathrm{Tor}_n^R(f, N) = H_n(\mathrm{id} \otimes f).$$

To show that Tor is a bifunctor, we show it is functorial in both coordinates, and that they commute.

Firstly, Tor is functorial in its first coordinate since given maps $M \xrightarrow{f} M' \xrightarrow{f'} M''$, the induced maps $\mathrm{Tor}^R(M, N) \to \mathrm{Tor}^R(M', N) \to \mathrm{Tor}^R(M'', N)$ are the quotients of $\mathrm{id} \otimes f$ and $\mathrm{id} \otimes f'$ respectively. Quotienting is functorial, and $\mathrm{id} \otimes (f' \circ f) = (\mathrm{id} \otimes f') \circ (\mathrm{id} \otimes f)$, so $\mathrm{Tor}^R(M, N) \to \mathrm{Tor}^R(M'', N)$ is equal to the composition of the maps. And the identity $M \xrightarrow{\mathrm{id}} M$ induces $\mathrm{Tor}^R(M, N) \to \mathrm{Tor}^R(M, N)$ given by quotienting $\mathrm{id} \otimes \mathrm{id} = \mathrm{id}$, which is again the identity.

And Tor is functorial in the second coordinate since if we have $N \xrightarrow{f} N' \xrightarrow{f'} N''$, with free resolutions $\mathscr{R}, \mathscr{R}', \mathscr{R}''$, then f lifts to $F \colon \mathscr{R} \to \mathscr{R}'$ and f' lifts to $F' \colon \mathscr{R}' \to \mathscr{R}''$. Clearly then $F' \circ F \colon \mathscr{R} \to \mathscr{R}''$ is a lift of $f' \circ f$, and so $\mathrm{Tor}(M, f' \circ f)$ is the quotient of $F' \circ F$. Quotienting is functorial, so this is equal to the quotient of F' composed with the quotient of F, so $\mathrm{Tor}(M, f') \circ \mathrm{Tor}(M, f)$ as desired. And $\mathrm{id} \colon \mathscr{R} \to \mathscr{R}$ is a lift of $\mathrm{id} \colon N \to N$ since the identity is always a chain complex morphism, and quotienting the identity gives the identity. Thus $\mathrm{Tor}(M, \mathrm{id}) = \mathrm{id}$ as desired.

Now suppose we have $f \colon M \to M'$ and $g \colon N \to N'$. Let $f^* = \mathrm{Tor}(f, \bullet)$ and $g_* = \mathrm{Tor}(\bullet, g)$, so we want to show

$$\begin{tikzcd}
\mathrm{Tor}(M, N) \arrow[r, "f^*"'] \arrow[d, "g_*"'] & \mathrm{Tor}(M', N) \arrow[d, "g_*"'] \\
\mathrm{Tor}(M, N') \arrow[r, "f^*"] & \mathrm{Tor}(M', N')
\end{tikzcd}$$

commutes. Let $\mathscr{R}, \mathscr{R}'$ be free resolutions of N, N' respectively, and denote the lift of $g \colon N \to N'$ by $G \colon \mathscr{R} \to \mathscr{R}'$. Then

$$\mathrm{Tor}_n(M, g) = H_n(G \otimes \mathrm{id}_M), \qquad \mathrm{Tor}_n(f, N) = H_n(\mathrm{id}_{N_n} \otimes f).$$

So we want to show

$$H_n(\mathrm{id}_{N'_n} \otimes f) \circ H_n(G \otimes \mathrm{id}_M) = H_n(G \otimes \mathrm{id}_{M'}) \circ H_n(\mathrm{id}_{N_n} \otimes f).$$

Both sides are equal to $H_n(G \circ f)$ as desired.

Thus Tor is indeed a bifunctor.

(11) Let $0 \to N \to K \to L \to 0$ be a SES. Let $\mathscr{R} \to M$ be a free resolution, and since free modules are flat we get a SES of chain complexes $0 \to \mathscr{R} \otimes N \to \mathscr{R} \otimes K \to \mathscr{R} \otimes L \to 0$. Homology gives long exact sequences from short exact sequences of complexes (via the snake lemma), so this gives a long exact sequence

$$\cdots \to H_{i+1}(\mathscr{R} \otimes L) \to H_i(\mathscr{R} \otimes N) \to H_i(\mathscr{R} \otimes K) \to H_i(\mathscr{R} \otimes L) \to H_{i-1}(\mathscr{R} \otimes N) \to \cdots.$$

Since $\mathrm{Tor}_i^R(\bullet, M)$ is defined to be $H_i(\mathscr{R} \otimes \bullet)$, this is a long exact sequence

$$\cdots \to \mathrm{Tor}_{i+1}^R(L, M) \to \mathrm{Tor}_i^R(N, M) \to \mathrm{Tor}_i^R(K, M) \to \mathrm{Tor}_i^R(L, M) \to \mathrm{Tor}_{i-1}^R(N, M) \to \cdots.$$

This ends with $M \otimes_R N \to M \otimes_R K \to M \otimes_R L \to 0$ because $\mathrm{Tor}_0^R(M, N) = M \otimes_R N$.

(12) Let N be free, then $0 \to N \to N \to 0$ is a free resolution of N, and so $\mathrm{Tor}(M, N)$ is the homology of $0 \to M \otimes N \to 0$. This has zeroth homology isomorphic to $M \otimes N$, and all other homology modules are zero, as desired.

(13) If M is flat then $\bullet \otimes_M M$ is exact. Thus if $\mathscr{R} \to N \to 0$ is exact then so is $\mathscr{R} \otimes M \to N \otimes M \to 0$. So the chain complex $\mathscr{R} \otimes M \to 0$ is exact at all indices greater than zero and thus its homology modules, $\mathrm{Tor}_i^R(M, N)$, are zero.

Conversely if $\mathrm{Tor}_1^R(M, N) = 0$ for all N, then let $0 \to L \to K \to N \to 0$ be SES, so we have a long exact sequence

$$\cdots \to \mathrm{Tor}_1^R(M, N) = 0 \to M \otimes L \to M \otimes K \to M \otimes N \to 0,$$

so tensoring with M maps a SES to a SES, and thus M is flat.